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Rapid Flood Inundation Mapping for Effective Management: A Machine Learning and Pixel-Based Classification Approach in Feni District, Bangladesh

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ABSTRACT

In Bangladesh, where floods frequently occur, there is a severe annual risk to community displacement, agriculture, fisheries, livestock, public health, and food security. Extreme flooding events are becoming more common due to a combination of human-induced climate change, irregular upstream river water flows, increased proportion of sediment distribution on the riverbed, institutional fragility, lack of planning regulations, and changing rainfall patterns. Effective flood management requires precise and timely flood mapping methodologies to adopt disaster risk reduction strategies and enable efficient response efforts. This study presents an approach that facilitates timely flood identification, improving emergency response, evacuation initiatives, relief distribution, and disaster risk reduction. This research introduces a novel methodology for expedited flood inundation mapping, using the August 2022, 2023, and especially the 2024 flood events in the Feni District as the primary case study. The study employs Google Earth Engine (GEE) and Sentinel-1 synthetic aperture radar (SAR) data to accurately delineate flood inundation regions by utilizing vertical transmit and vertical receive (VV), vertical transmit and horizontal receive (VH), and VV/VH polarization bands. Water bodies characterized by lower backscatter values in VH polarization ranging from −41.15 to −24.06 dB and VV polarization from −31.66 to −15.94 dB were identified as suitable thresholds for flood inundation area delineation. To assess the accuracy of flood map, this study focuses on pixel-based digital classification and machine learning (ML) techniques separately for flood inundation mapping. The classification accuracy values of 95.60% for the pixel-based method and 94.40% for the random forest ML model specifically correspond to the 2024 flood event. This study developed a GEE-based operational methodology by evaluating two innovative techniques designed for rapid flood inundation mapping to support effective flood management and disaster risk reduction efforts.

1 | Introduction

Floods are one of the most devastating natural hazards in Bangladesh that impacting millions of people each year (Islam, Abdullah, et al. 2024; Rasid and Pramanik 1993; Uddin et al. 2021). The country's vulnerability stems from its location in the downstream Ganges–Brahmaputra–Meghna (GBM) delta, where heavy monsoon rains, upstream river overflows, and inadequate disaster management systems contribute to recurrent flooding. Flood events cause widespread destruction, displacing communities, damaging agriculture, fisheries, and infrastructure, and disrupting essential services such as education and healthcare (Hossain et al. 2020; Pizzorni et al. 2024). Agriculture, the backbone of the economy, is particularly affected by regular inundation, intensifying food insecurity, poverty, and economic instability

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(Hossain et al. 2020; Parven et al. 2022). With climate change increasing the frequency and severity of floods and other anthropogenic factors worsening the situation (Fahad et al. 2024; Nones and Guo 2023; Tabari 2020; Thapa et al. 2024), there is an urgent need for an effective flood management system to protect critical infrastructure and vulnerable communities (Grigg 2024; Uddin and Matin 2021).

Beyond early warning systems and evacuation planning, a rapid and scalable flood inundation mapping methodology is essential for effective disaster response. Such a system would support first responders, aid workers, and policymakers in making timely decisions about evacuation, resource distribution, and emergency relief. Advances in geospatial technologies, including geographic information systems (GIS), remote sensing, and artificial intelligence (AI), have made near-real-time flood mapping increasingly feasible (Haque et al. 2019; Rambhad et al. 2023). Satellite-based remote sensing, particularly synthetic-aperture radar (SAR), is crucial for continuous flood monitoring, especially in inaccessible or hazardous regions.

Among the Earth observation satellites in orbit, Sentinel-1 synthetic aperture radar (SAR) sensors provide all-weather, day-and-night imaging at no cost (Amitrano et al. 2024; Ghosh et al. 2022). Integrating SAR data with machine learning (ML) algorithms like random forest (RF) on platforms such as Google Earth Engine (GEE) significantly enhances the efficiency and automation of flood mapping (Ghosh et al. 2023; Mayer et al. 2025; Uddin et al. 2021). These approaches allow large datasets to be processed rapidly, distinguishing flooded areas with high accuracy without the need for extensive manual data retrieval (Liao et al. 2023; Refice et al. 2018). The automation of flood mapping through GEE scripting enables real-time monitoring, making it an invaluable tool for flood response and management. Additionally, SAR-based digital classification methods refine flood mapping precision by leveraging backscatter measurements, reducing processing time while improving accuracy.

Although several ML algorithms can be used for flood detection, RF has emerged as one of the most effective methods. Its ability to handle high-dimensional datasets and manage both continuous and categorical variables makes it particularly well-suited for SAR-based classification. Unlike other algorithms, RF can also assess the importance of different input variables, such as VV and VH polarization values, which contribute to more accurate flood inundation mapping. This makes RF a preferred choice for applications requiring high classification accuracy, such as flood mapping in challenging environments like Bangladesh.

These technological advancements can significantly strengthen flood management efforts in Bangladesh by optimizing emergency response, evacuation planning, and post-disaster recovery. This study demonstrates how SAR data, GEE, and classification may speed up flood mapping and improve resilience. Continuous flood inundation mapping also supports long-term resilience planning by identifying high-risk areas and guiding infrastructure investments, such as flood shelters, to mitigate future risks (Phongsapan et al. 2019). Against this backdrop, this study develops an operational methodology for rapid flood inundation mapping, using the August 2024 flood event as the main case study, with additional historical assessment for August 2022 and 2023 to capture flood recurrence and trends. By leveraging Sentinel-1 SAR imagery and ML techniques on the GEE platform, the study aims to improve flood detection accuracy through a comparative analysis of pixel-based and ML-based classification techniques. To ensure the methodology is adaptable across different geographic contexts by establishing a systematic and transferable framework, this study contributes to improving real-time flood monitoring and decision-making, supporting more effective disaster response and mitigation efforts. The study follows a structured operational framework as described below:

a. SAR image preprocessing, including thermal noise removal, speckle filtering, and feature extraction based on VV and VH polarization backscatter values.
b. Classification and accuracy assessment, applying pixel-based and RF classification using predefined training and validation samples.
c. Scalability, ensuring the approach can be applied to various flood events in different locations with minor adjustments.
d. Flood progression analysis, capturing spatial and temporal changes in flood extent to enhance long-term flood management strategies.

2 | Study Area

The Feni District in Bangladesh, which experienced flash floods, has been selected as a case study. Specifically, the August 2024 flooding event, triggered by heavy rainfall and upstream water flows, devastated communities across 11 districts in the eastern regions of Bangladesh, particularly in Feni, Comilla, Noakhali, Laxmipur, and Moulvibazar districts. According to the United Nations (UN) Country Team in Bangladesh, the flash floods affected over 5.8 million people, caused 52 deaths, disrupted services for over 1 million people, displaced 0.5 million individuals, damaged over 300,000 ha of crops, forcibly closed more than 7000 schools affecting over 1.75 million primary students, and inflicted losses to fisheries and livestock estimated at USD 122 million and USD 34 million, respectively (UN 2024).

In particular, Feni is selected for further analysis, as this district was severely impacted with over 150,000 people being displaced (ECHO 2024). The study area, shown in Figure 1, indicates that Feni District, located in the southeastern region of Bangladesh, spans latitude 22°44′–23°17′ and longitude 90°15′–91°35′. It covers an area of approximately 990 km^2 and is bordered by the Comilla district of Bangladesh and the Indian state of Tripura to the north. The district comprises six administrative sub-districts, officially called Upazilas: Chhagalnaiya, Daganbhuiya, Feni Sadar, Fulgazi, Parshuram, and Sonagazi. Approximately 32% of the district's total land area comprises urban communities, whereas the remaining 68% is rural. Agriculture is the principal source of income for 141,564 out of the 322,772 families. Several rivers, including the Feni, Muhuri, Kohua, Selonia, Kalidas, and Pahalia traverse the Feni District. These rivers remain navigable throughout the year and essential for water transportation and irrigation. Feni District receives an annual average precipitation of 2603 mm (BBS 2022).

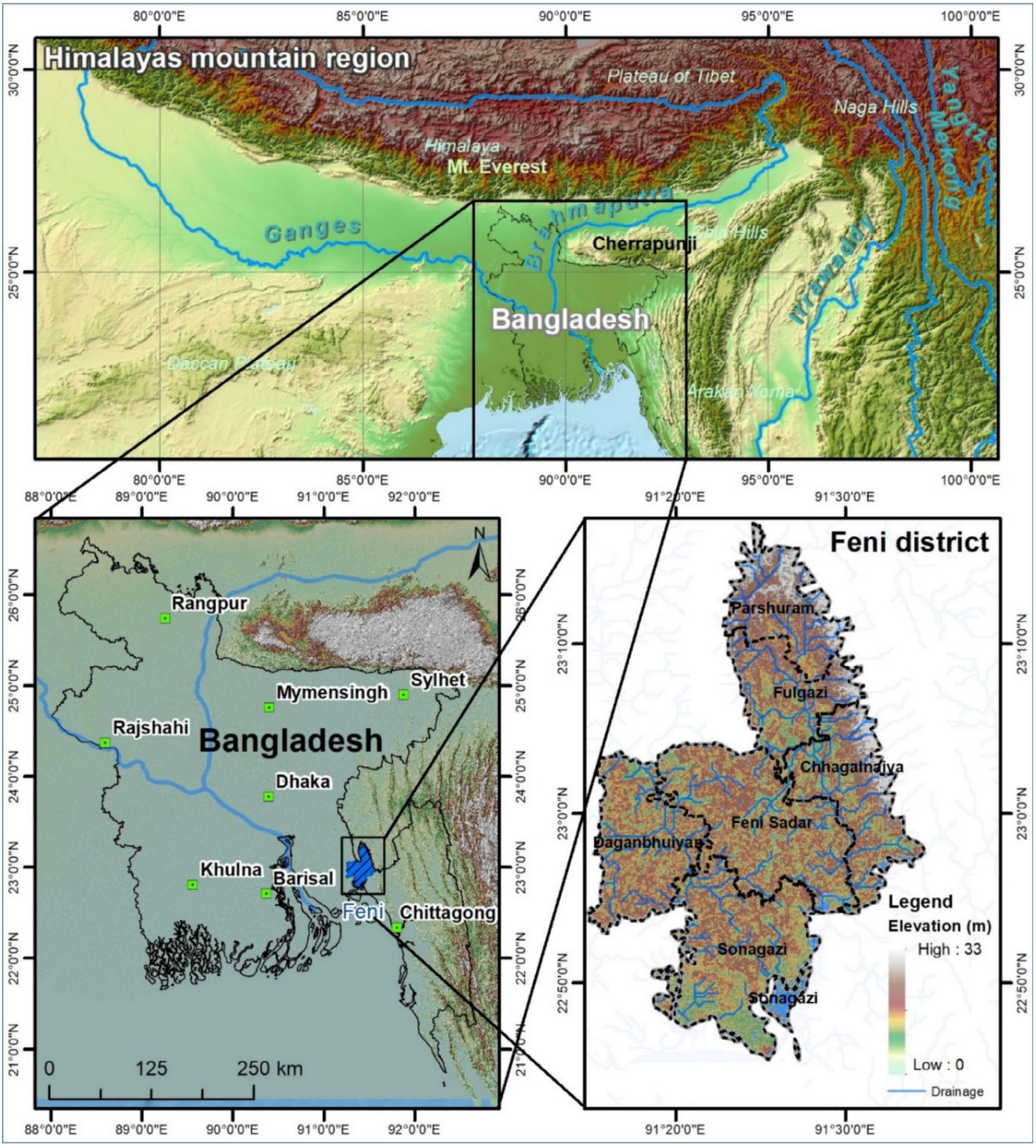


FIGURE 1 | The study area map—Feni District, Bangladesh.

3 | Method

3.1 | Data Used

Sentinel-1 ground range detected (GRD) imagery, identified by the ID numbers 20240821T120417 and 20240821T120442 (dated August 21, 2024), was used for flood inundation mapping, algorithm development, and method comparison. To establish a baseline for comparison, Sentinel-1 data from the dry season (January–March) was also analyzed to classify perennial water bodies. The SAR system of Sentinel-1, functioning in the C-band, is exceptionally effective in identifying surface water in all kinds of weather circumstances, including the overcast and wet situations commonly linked to flood occurrences. The GRD results underwent preprocessing to eliminate thermal noise to enhance spatial accuracy. The backscatter data in VV and VH polarizations differentiate waterlogged areas from other terrestrial characteristics (Figure 2). Water surfaces display less backscatter and appear dark in radar imaging, facilitating the identification of inundated areas.

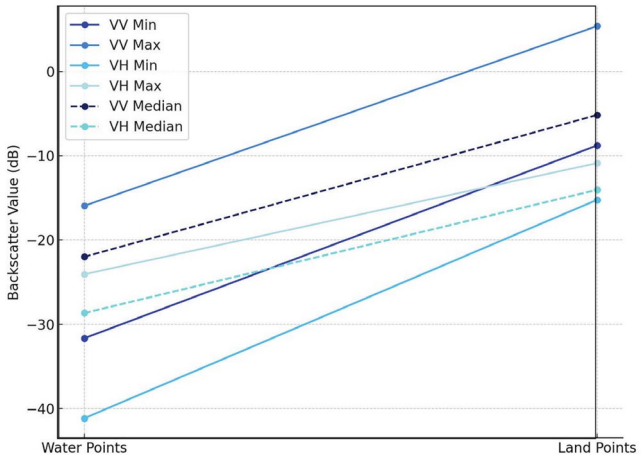


FIGURE 2 | Sentinel-1 SAR backscatter characteristics for flood inundation (water) and non-flooded (land) sample points in the 2024 Feni District flood. It displays minimum, maximum, and median VV and VH polarization backscatter values, separating flood and non-flooded areas.

Additionally, historical flood mapping data from August 2022 to 2023 were incorporated to analyze trends, determine recurring flood-prone areas, and identify newly inundated locations. This comprehensive approach provides a strong foundation for understanding flood patterns over multiple years and supports effective flood management strategies. Furthermore, OpenStreetMap (OSM 2021) data was used to assess flood exposure on infrastructure, specifically to determine the number of buildings affected by floodwaters. Integrating these datasets provides valuable support for flood management, enabling a better understanding of flood impacts and aiding in developing targeted mitigation strategies.

3.2 | Flood Inundation Mapping

Sentinel-1 data was processed on GEE to map the flood inundation area that went through a few pre-processing steps for SAR imagery analysis. The preprocessing procedures were thermal noise elimination, radiometric calibration, terrain correction utilizing SRTM 30 m DEM, and the implementation of the Lee speckle filter. The backscatter coefficient (σ°) was computed in decibels (dB) for each pixel, with a speckle filter applied to reduce granular noise and enhance feature visibility. To classify flood-inundated areas, a total of 320 reference points were manually digitized in GEE to identify flood-inundated areas, utilizing high-resolution imagery, which included 130 flood points, 130 land points, and 60 perennial water bodies. A randomly generated independent set of 250 validation sites was created across the study area to provide statistical independence and spatial representativeness. These reference points were used to extract VV and VH backscatter values from the August 21, 2024 Sentinel-1 image (Figure 2). The extracted values were then analyzed to establish the statistical differences between water and non-water surfaces, forming the basis for classification.

Flooded and non-flooded surfaces exhibit distinct SAR backscatter characteristics, which were utilized for mapping. Calm, open water bodies, such as lakes, reservoirs, and large rivers, experience specular reflection, where most of the radar energy is reflected away from the sensor. As a result, VV and VH backscatter values are very low, indicating minimal return signal. In contrast, partially submerged vegetation within floodwaters can create double-bounce scattering, where the radar signal reflects between the water surface and plant stems before returning to the sensor. This effect is more pronounced in VV polarization, sometimes resulting in elevated backscatter values in flooded forests, wetlands, or areas with emergent crops, making them appear brighter in SAR images than open water. However, even in these cases, backscatter values remain lower than those of dry land.

Using these backscatter differences, two classification methods were applied: pixel-based thresholding and ML-based classification. For pixel-based classification, threshold values were directly applied to the extracted VV and VH backscatter data to delineate flooded areas. In parallel, a RF is one of many ML algorithms, specifically an ensemble learning method based on a decision trees classifier (Pal 2005) was trained using the same set of 320 reference points. The RF model was developed using decision trees, where each tree was trained on a subset of the reference points. The model utilized VV and VH backscatter values as predictor variables, aiming to distinguish flood-inundated areas from land and permanent water bodies. The RF classifier learned from variations in backscatter values, identifying key thresholds that separated flooded pixels from non-flooded ones.

Once the classification was completed, the flood map was validated using an independent set of 250 reference points. Accuracy was assessed through confusion matrix analysis, overall accuracy, and the Kappa coefficient, and a detailed report on accuracy assessment is presented in Data S1. The final flood map was then used to calculate the flood-inundated area. To assess flood recurrence, historical Sentinel-1 data from August 2022 to 2023 were analyzed in ArcGIS Desktop to classify persistent flood-prone areas and newly inundated locations.

4 | Results

The flood inundation map yielded some promising results, employing two methodologies a pixel-based digital classification and a ML-based RF approach. Both utilized the VV, VH, and VV/VH polarization bands to delineate the extent of flood inundation.

The backscatter analysis of both land and water sample points using Sentinel-1 radar data reveals distinct differences in their interaction with VV and VH polarizations, which is crucial for accurate flood inundation mapping. Water bodies exhibit significantly lower backscatter values than land, particularly in the VH polarization, where values range from −41.15 to −24.06 dB, with an average of −31.29 dB. In the VV polarization, water points reflect slightly higher radar energy, ranging from −31.66 to −15.94 dB with an average of −23.20 dB (Figure 2). These lower backscatter values are attributed to water's smooth, reflective surface, which reflects radar waves away from the sensor,

TABLE 1 | Error matrix for the pixel based flood inundation map of August 2024.

Class name	Reference total	Classified total	Number correct	Producers' accuracy	Users' accuracy
Flood inundation	95	86	85	89.47%	98.84%
Perennial water	17	19	17	100.00%	89.47%
Other	138	145	137	99.28%	94.48%
Total	250	250	239		

Note: Overall classification accuracy = 95.60% and overall Kappa statistics = 0.9191.

TABLE 2 | Error matrix for the RF-based flood inundation map of August 2024.

Class name	Reference total	Classified total	Number correct	Producers' accuracy	Users' accuracy
Flood inundation	95	95	88	92.63%	92.63%
Perennial water	17	14	12	70.59%	85.71%
Other	138	141	136	98.55%	96.45%
Total	250	250	236		

Note: Overall classification accuracy = 94.40% and overall Kappa statistics = 0.8964.

resulting in weak returns. This distinct difference in backscatter values between water and land, especially in the VH polarization, significantly enhances flood map accuracy. The accuracies of users and producers are displayed in Tables 1 and 2.

Conversely, land points demonstrate higher backscatter values due to their rough and complex surfaces, including vegetation, buildings, and soil. VV polarization for land points ranges from −8.78 to 5.39 dB, with an average of −2.85 dB, reflecting more substantial radar returns caused by interactions with vertical structures. The VH polarization for land, although lower than VV, still exhibits higher backscatter values than water, ranging from −15.23 to −10.87 dB, with an average of −13.38 dB, indicating increased scattering from complex surfaces such as vegetation and uneven terrain (Figure 2).

For the accuracy assessment of the flood maps, 250 reference points were randomly generated for validation of both the pixel-based digital classification and the ML methods. The pixel-based classified map shows an overall classification accuracy of 95.60%, with an overall Kappa statistic of 0.9191. This is an indication of a high level of agreement. In comparison, the ML-based RF model produced an overall classification accuracy of 94.40%, with an overall Kappa statistic of 0.8964. The validation results demonstrate slightly lower performance than the pixel-based method; however, it is still a robust classification method. A detailed report on the accuracy assessments can be found in Tables 1 and 2, as well as GEE code S1.

The digital classification method identified a total flooded area of 201 km^2 (Figure 3b), whereas the ML approach detected a larger area of about 254 km^2 (Figure 3c). Marked discrepancies in flood inundation extent were noted across the Upazilas/sub-districts. Sonagazi encountered the most flooded areas, with 49 km^2 determined using the pixel-based digital classification and 60 km^2 with the ML method, respectively. Daganbhuiyan exhibited the least inundated areas, calculating 16 km^2 per the digital classification method and 23 km^2 according to the ML method, respectively. The ML approach consistently calculated larger flooded areas, capturing a more extensive flood extent across the study area. For example, 20% of the district was mapped as inundated by flood water using the digital classification method, whereas 26% of areas were identified as inundated using the RF-assisted ML method. Finally, the flood maps generated in this study were shared with the relevant disaster management organizations in Bangladesh, including Sentinel Asia (AentinelAsia 2024). Any user can publicly access the GEE codes used in this study and perform flood mapping from this link GEE code (see the Data S2).

This study emphasizes the superior ability of ML to precisely delineate the comprehensive scope of flooding, demonstrating its utility as an effective instrument for expedited flood mapping—particularly, digital classification methods yielded maps with increased precision. Integrating SAR data with ML models, such as RF, presents considerable potential for enhancing flood catastrophe management and mitigation in flood-prone areas like Bangladesh.

The flood inundation maps for August for the years 2022, 2023, and 2024 are shown in Figures 4 and 5. The analysis of sub-district (Upazial) wise flood inundation areas from 2022 to 2024 in Feni District reveals a significant increase in flood-affected areas over the years. In 2022, the total flood inundation area was 27 km^2, which slightly decreased to 17 km^2 in 2023. However, 2024 witnessed a catastrophic flood, with 201 km^2 inundated. Among the Upazila, Sonagazi experienced the maximum inundation in 2024, covering 49 km^2, followed by Chhagalnaiya (37 km^2), Fulgazi (35 km^2), and Feni Sadar (33 km^2). The lowest inundation was mapped in Daganbhuiyan (16 km^2) and Parshuram (31 km^2). The severity in flood-affected areas in 2024 suggests a growing vulnerability, necessitating urgent flood mitigation measures, adaptive land-use strategies, and flood forecasting with the support of upstream discharge information.

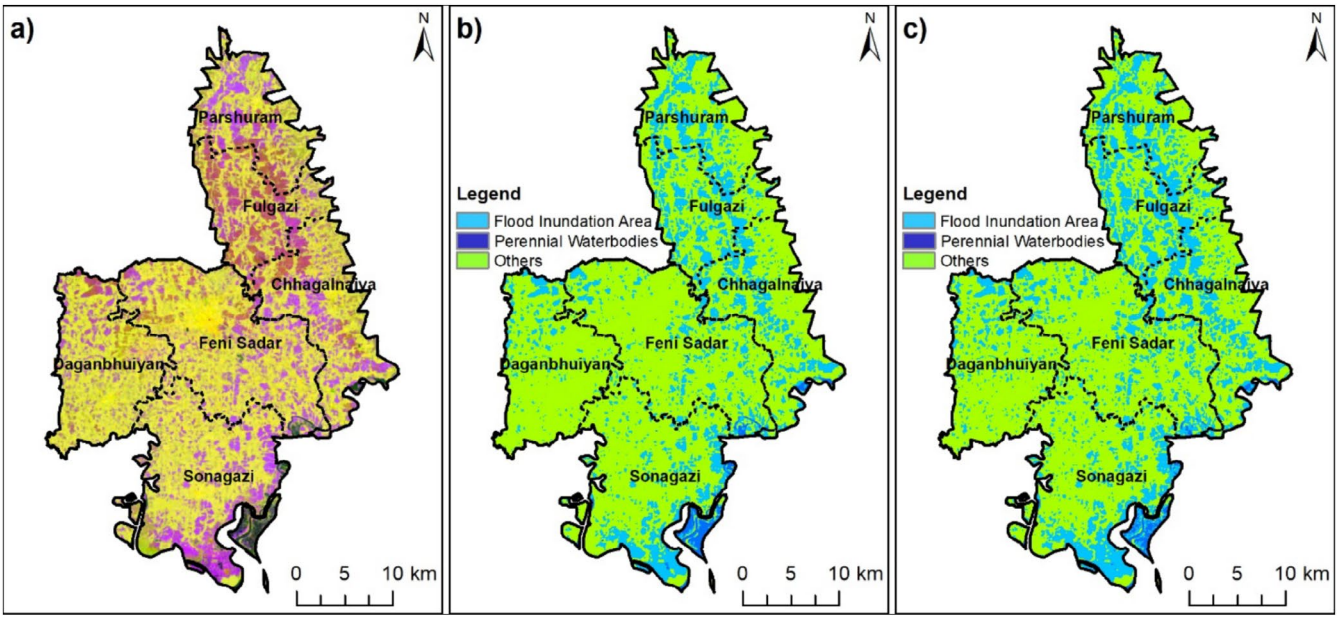


FIGURE 3 | Flood inundation maps of Feni District, Bangladesh (a) Sentinel-1 composite showing flooded areas appear in pink, perennial water bodies in black, and land areas in yellow; (b) flood inundation map produced by the pixel-based classification; and (c) flood inundation map produced by the random forest ML algorithm.

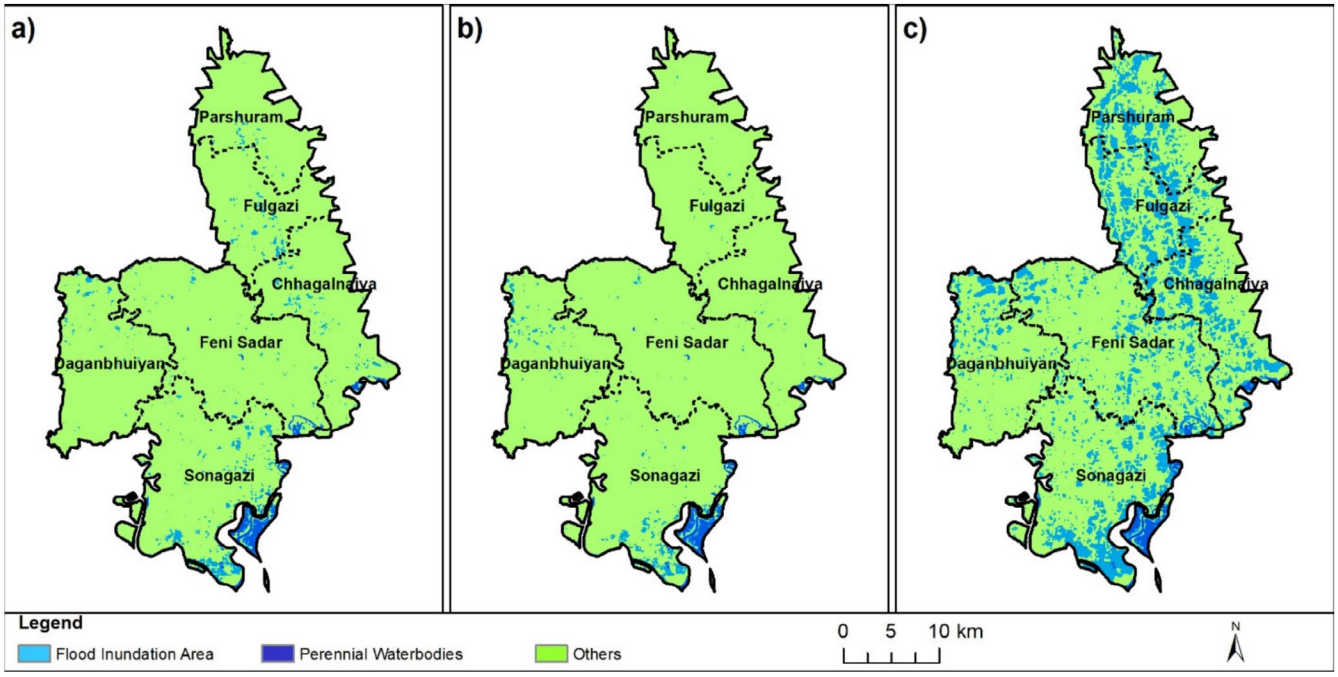


FIGURE 4 | Complete flood inundation map for Feni District, Bangladesh for the months of (a) August 2022, (b) August 2023, and (c) and August 2024.

The flood recession and rise patterns in Feni, Bangladesh, for August 2022, 2023, and 2024 exhibit notable spatial variations across Feni District. According to Figure 6, a total of 18 km^2 was inundated in August 2022, with 9 km^2 remaining commonly inundated in both years. However, an additional 8 km^2 of new areas were inundated in August 2023, indicating an expansion in flood extent. Sonagazi experienced the highest flood exposure among the Upazilas, with 7 km^2 commonly inundated and 4 km^2 newly flooded in 2023. Similarly, Fulgazi and Chhagalnaiya saw 3 km^2 of newly affected areas, highlighting shifts in flood-prone zones. Daganbhuiyan recorded a significant increase in new inundations with 3 km^2, whereas Parshuram had minimal changes.

The flood patterns between August 2023 and August 2024 indicate a significant expansion in inundated areas across Feni, Bangladesh. The commonly inundated regions have increased, with Sonagazi Upazila experiencing the most widespread and

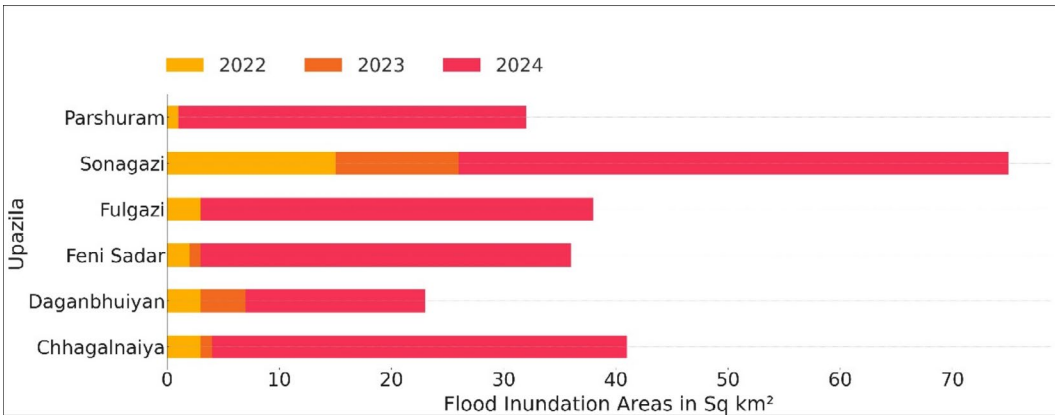


FIGURE 5 | Upazila-wise flood inundation areas for August 2022, 2023, and 2024.

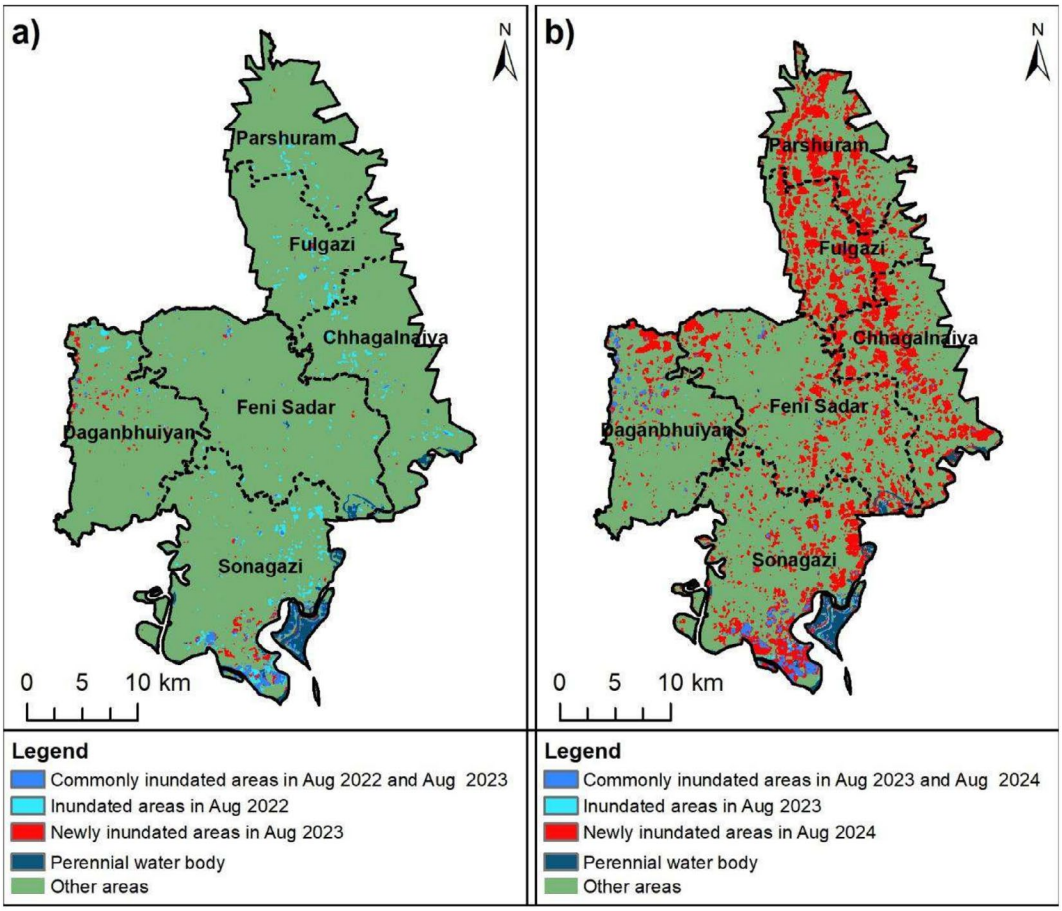


FIGURE 6 | The flood recession and rise patterns in Feni, Bangladesh, highlighting spatial changes between (a) 2022 and 2023 and (b) 2023 and 2024.

persistent flooding at 90.72 km^2, followed by Daganbhuiyan (34.03 km^2) and Feni Sadar (9.12 km^2). Newly inundated areas in August 2023 were relatively small in most districts, with Sonagazi (4.53 km^2) and Feni Sadar (1.12 km^2) showing moderate increases.

Additionally, OpenStreetMap building layer data were integrated to estimate the number of buildings that were exposed to the flood, with a small portion of the area visualized in 3D in Figure 7. The table presents data on the number of buildings and those exposed to floods in different upazilas of Feni District, Bangladesh (Data S3). Among the listed upazilas, Feni Sadar has the highest number of buildings (56,274), whereas Sonagazi has the highest number of flood-exposed buildings (311). Parshuram and Fulgazi also show significant exposure, with 247 and 179 buildings affected, respectively. In contrast, Daganbhuiyan has the lowest flood exposure, with only 12 buildings at risk. This data highlights the varying degrees of flood vulnerability across different upazilas, emphasizing the need for targeted flood mitigation efforts in the most affected areas.

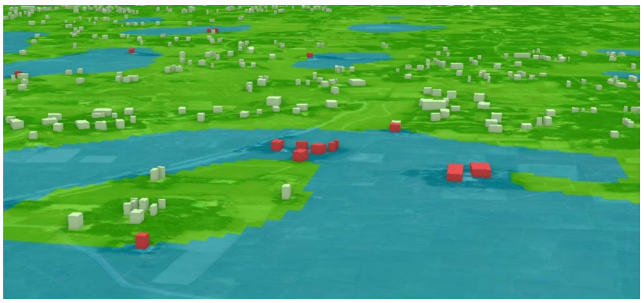


FIGURE 7 | Visualization of flood-exposed buildings, with a selected portion of the area presented in 3D.

5 | Discussion

The results of this study underscore the effectiveness of both traditional pixel-based classification and ML techniques, specifically the RF algorithm, for rapidly mapping flood inundation in Bangladesh. Utilizing Sentinel-1 SAR imagery on the GEE platform, both methods achieved high accuracy in identifying flood-affected areas, with the pixel-based approach slightly outperforming the RF approach. These results emphasize the value of pixel-based classification methods with SAR satellite data for real-time flood monitoring in highly vulnerable areas such as the Feni District and other parts of Bangladesh for operational flood mapping. Moreover, RF-based flood inundation mapping also has potential in Bangladesh as it produces high accuracy.

Given Bangladesh's exposure to frequent heavy rainfall, between 1500 and 4000 mm of rainfall per year (Islam and Nursey-Bray 2017; Rahman et al. 2018)—automatically mapping flood inundation using SAR data is particularly beneficial (Raju et al. 2022; Uddin and Matin 2021). The capability to produce maps with higher accuracy is crucial for timely response efforts, mainly when communication networks are down, providing rescue teams with reliable information to prioritize efforts in affected communities. Pixel-based digital classification is comparatively quicker as we do not need to insert training samples from other sources (Balha et al. 2021; Refice et al. 2022). Floodwater and inundated landscapes also make it challenging to collect reference data from the ground (Schumann et al. 2023).

The use of Sentinel-1 SAR imagery on the GEE platform significantly enhanced flood mapping in Feni District by enabling real-time processing and classification of flood-affected areas. The combination of VV and VH polarization bands provided a clear distinction between water and land, improving flood detection accuracy. GEE's cloud computing environment facilitated large-scale image analysis without requiring extensive local computational resources, making it ideal for rapid response efforts. Moreover, the pre-processing capabilities in GEE, including speckle filtering, thermal noise removal, and geolocation correction, ensured that the Sentinel-1 images were optimized for accurate classification. The seamless integration of SAR data and automated classification methods in GEE enabled quick and efficient flood inundation mapping, providing essential data for disaster response teams and policymakers.

Although drones can be used for high-resolution flood mapping, their limitations must be acknowledged. Drones are suited for small, localized areas but are less practical for large-scale flood zones, where rapid and extensive coverage is essential (Iqbal et al. 2023). Drones are also not useful on rainy days. Additionally, drones generate vast amounts of data requiring desktop-based processing, potentially delaying the creation of timely maps necessary for relief operations. Therefore, although drones can be valuable for detailed local surveys or media coverage, they are not ideal for broad, large-scale flood mapping efforts (Singh et al. 2024).

Although much of the existing literature focuses on the accuracy of flood maps, their reliability and quality are equally crucial for effective disaster management. Authentication of flood maps during an active event poses significant challenges, because access to inundated areas is often tricky (Jerolleman 2020; Mudashiru et al. 2021). This underscores the importance of strategic resource allocation for ground-based validation and developing alternative systems to ensure map reliability in real-time scenarios. Flood risk assessment should be integrated with vulnerability indicators such as population density, socio-economic conditions, and proximity to flood-prone areas to identify high-risk zones effectively (Chen et al. 2021; Mayer et al. 2025).

In this context, enhanced by innovative band combinations, SAR image analysis is pivotal in generating high-quality flood maps (Amitrano et al. 2024; Luo and Pi 2023). A better way of processing SAR data, integrated with optical imagery, can enhance classification accuracy by providing more precise backscatter values, which improve flood detection. These refined backscatter values, once determined, can be reused for similar locations and conditions in future years. However, recalibration of threshold values is necessary when the location or SAR data changes to ensure continued mapping accuracy. The RF-based classification technique relies on accurately collected water and land training samples. The mobility of flood water, which is never static, means that training samples collected from previous years may not yield high-quality maps.

Following the flood inundation mapping conducted over multiple years, the data can be leveraged for comprehensive flood risk assessments and to identify suitable locations for flood shelters. During prolonged flood events, flood shelters become critical for effective disaster management, providing safe places for affected communities (Hasan et al. 2024; Uddin and Matin 2021). The suitability of flood shelter locations can be determined through GIS analysis, integrating flood maps, land cover data, and additional layers such as elevation, proximity to settlements, and accessibility (Mayer et al. 2025). This approach ensures that shelters are strategically placed in areas where they are most needed, minimizing risks and enhancing the resilience of vulnerable communities during flood events (Lee et al. 2021; Mabahwi et al. 2021).

Though for this study's exposure assessment purpose, particularly for buildings, we conducted an analysis by overlaying the flood maps with OpenStreetMap buildings. However, agencies such as ICIMOD, ESRI, or Copernicus produce land cover datasets that can also be used for crop damage estimation. Integrating flood maps with land cover data enables authorities to evaluate impacts on various land uses, including agriculture, forests, and urban areas. This integration is vital for estimating

crop losses and infrastructure damage, informing post-disaster recovery planning, and shaping compensation schemes (Haque and Jahan 2015; Islam, Ahamed, et al. 2024).

It is crucial to recognize that the accuracy of these damage assessments can also be influenced by factors such as flood duration and inundation depth (Bofana et al. 2022; Cohen et al. 2018; Uddin et al. 2019). More prolonged or deeper flooding can lead to more considerable damage to both crops and infrastructure. Therefore, remote sensing-based studies may focus on field-based damage assessment. Sometimes, many buildings may be inundated, but they should not be considered damaged. The study highlights the advantages of SAR-based mapping while recognizing its limitations in heavily forested and urban environments, which are attributed to signal scattering and delay effects. To enhance mapping accuracy, subsequent efforts should explore the integration of SAR with optical imaging and additional geographic layers. Thus, incorporating flood characteristics—such as water depth and duration—into automated mapping and damage estimation models will enhance their reliability. Ultimately, integrating these elements into the flood mapping and damage estimation processes will strengthen disaster response and recovery strategies, providing a more effective tool for managing the impacts of recurrent flooding in Bangladesh. Furthermore, a multi-criteria approach combining hazard intensity, exposure, and vulnerability assessments can improve decision-making for flood risk management, ensuring that mitigation efforts are directed toward the most at-risk communities.

6 | Conclusion

This research highlights the efficacy of integrating Sentinel-1 SAR data with pixel-based classification and ML methods for expedited flood mapping in the Feni District during the August 2024 floods. We effectively processed large datasets by employing GEE, producing precise flood maps timely. Consequently, the established approaches may be employed for operational flood mapping objectives. The pixel-based technique and the RF classifier exhibited great accuracy, with the pixel-based method outperforming in complicated scenarios reliant on reference points. This type of mapping is remarkably effective for real-time flood mapping and provides substantial benefits for disaster management in flood-prone regions. Although the pixel-based classifier exhibits superior performance, the ML methodology remains valuable. The research confirms the efficacy of these techniques for real-time flood monitoring and underscores the importance of dependable training samples and flexible thresholds. However, the RF classifier is sensitive to training data quality and may struggle with misclassification in highly heterogeneous flood landscapes, particularly when reference data are limited. These findings highlight the critical importance of satellite imagery and ML in improving flood response and management tactics in Bangladesh, a nation with heightened risks from climate change-induced flooding. Further research into optimizing flood mapping techniques, including refining classification algorithms and enhancing data integration methods, is crucial for improving accuracy and reliability. The effectiveness of operational flood mapping initiatives depends on their technical robustness. Advanced technological tools and enhanced community engagement are essential for strengthening flood response systems. A comprehensive research agenda is crucial to optimize the efficacy of flood response and management at high-risk areas.

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Ethics Statement

We confirm that this manuscript is original, has not been published elsewhere, and is not under consideration for publication elsewhere. We also declare no conflicts of interest related to this study. This paper not involved in any human and animal related experimental research.

Conflicts of Interest

The views and interpretations in this publication are those of the authors. They are not necessarily attributable to ICIMOD and do not imply the expression of any opinion by ICIMOD concerning the legal status of any country, territory, city or area or its authority, or concerning the delimitation of its frontiers or boundaries, or the endorsement of any product.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.